**Week 8: 19/10/2026 – 23/10/2026**

**Subject: Mathematics Course 3** **Unit Title: Exploring Transformations and Linear Relationships** **Grade : 8**

| Lessons الحصص | Learning targets أهداف التعلم | Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز | Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور |
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| **Lesson 1: Comparing Proportional Relationships** | I can compare proportional relationships. | 1.1 Flexible display of information (size, color, font, layout).<br>2.3 Support decoding of text and notation using tools, audio, and multiple representations.<br>8.1 Make goals clear, visible, and supported by short-term objectives. | MAGMA MATH:<br><br>Students will check the posted assignments /lessons on Magma<br><br>Girls 'campus: Students will check their materials on Files section on Teams. |
| **Lesson 2: Linear relationships** | I can identify linear and non-linear relationships. | | |
| **Lesson 3: Slope and similar triangles-Part 1** | I can understand how similar right triangles can be used to calculate and compare slope on a straight line. | | |
| **Lesson 4: Slope and similar triangles-Part 1** | I can justify why slope remains constant along a straight line using similar triangles and real-life UAE applications. | | |
| **Teacher's notes/ Reminders ملاحظات المعلم والتذكير** | Make sure your devices are fully charged.<br>Bring your pencil case/calculator with you.<br>Complete your extra practice on Magma Math App. | | |

| **Technology Integration** | Resources shared on Teams |
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| دمج التكنولوجيا | |
| **Cross-curricular links**<br>الروابط بين المواد الدراسية | Science Connection: Students explore how slope relates to the steepness of inclined planes and how it affects the force needed to move objects up a ramp.<br><br>IT Connection: Students use GeoGebra or Desmos to construct similar triangles on a straight line and investigate how the rise/run ratio remains constant. |
| **External Assessment Integration** دمج التقييمات الخارجية | All content is covered in MAP |
| **Wellbeing/ ATL Skills**<br>الصحة النفسية/ أساليب التعلم | 🧠 **Wellbeing**<br>• Confidence in solving real-world measurement problems.<br>• Resilience when learning from mistakes.<br>• Collaboration through shared investigations and discussion.<br>• Self-awareness by checking the reasonableness of solutions.<br>🌍 **IB ATL Skills**<br>**Thinking:** solving problems using scale factors and proportional reasoning.<br>**Communication:** explaining mathematical methods and solutions clearly.<br>**Research:** interpreting scale drawings and real-world models.<br>**Self-Management:** organising calculations accurately and efficiently.<br>**Social:** discussing strategies and comparing solutions with others |
| **Standards**<br>المعايير | 8.F.A.1: Understand that a function is a rule that assigns each input exactly one output.<br><br>8.F.B.4: Construct a function to model a linear relationship between two quantities and interpret the rate of change and initial value<br><br>MP3: Construct viable arguments and critique the reasoning of others. |
| **Inquiry/ TOK Question** | How does the accuracy of a scale factor affect the final size and reliability of a real-world design? |

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| **TOK البحث والاستقصاء أسئلة** | How can slope help us understand and compare rates of change in real-life situations?<br>How can a graph help us identify and understand the unit rate in a proportional relationship?<br>How can unit rates help us compare proportional relationships represented in different ways? |
| **Success Criteria معايير النجاح (Measurable)** | Lesson 1:<br>🟦 All students must be able to identify the unit rate in proportional relationships.<br>🔵 Most students should be able to compare proportional relationships from tables, graphs, equations, and descriptions.<br>🔺 Some students could be able to analyze and justify comparisons in real-world contexts, such as contractor rates, transport speeds, or energy outputs.<br>Lesson 2:<br>🟦 All students must be able to identify linear and non-linear relationships.<br>🔵 Most students should be able to find the value of y when x is given.<br>🔺 Some students could be able to solve geometry problems related to linear relationships.<br>Lesson 3:<br>🟦 All students must be able to identify the rise and run of right triangles formed along a straight line.<br>🔵 Most students should be able to calculate the slope of a line using the rise/run ratio of similar triangles.<br>🔺 Some students could be able to explain why different-sized similar triangles on the same line have equal slopes<br>Lesson 4:<br>🟦 All students must be able to identify similar triangles on a straight line and calculate their slopes.<br>🔵 Most students should be able to demonstrate that the slope remains constant by comparing the rise/run ratios of different similar triangles.<br>🔺 Some students could be able to justify why slope remains constant using triangle similarity and apply their reasoning to UAE ramps, roads, or rail routes. |
| **National Identity Domain مجالات الربط بالهوية الوطنية** | Belonging الإنتماء Respect الاحترام Heritage التراث |
| | Students connect slope and proportional relationships to **UAE roads, transport routes, and infrastructure**, strengthening their understanding of how mathematics supports their community. |

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| | Students demonstrate respect by **collaborating, sharing mathematical strategies, and appreciating different approaches** when solving UAE real-world problems.<br>Students explore proportional relationships through **UAE heritage contexts**, such as traditional architecture, falaj irrigation systems, and historical trade routes. |